Explanation of Individual Decisions



General Motivation

- Let's be clear, explanations are not intrinsically important for learning!
  - At least not human-interpretable explanations.

- A network can probably explain its decision to another network by just communicating its active weights to the other network!

- So,

explanations are essentially for interaction with humans



General Motivation

- But interaction with humans are of intrinsic value, some main reasons:
  - Trust in AI (AI-based medical diagnosis and prognosis)
  - Scientific (or generally knowledge) discovery from AI (ML for fluid dynamics)
  - Collaboration between Human and AI (sociable, cooperative AI)
  - …



General Motivation

- That being said,
  - Humans have evolved into current form of communication and rationalization
  - Learning through (self-)explanation (“deduction”) can be more efficient than learning through examples (“induction“)



Regulations and rules: “Right to explanation” demanded by users



“the existence of automated decision-making, including profiling, referred to in Article 22(1) and (4) and, at least in those cases, **meaningful information about the logic involved**, as well as the significance and the envisaged consequences of such processing for the data subject.”

[GDPR, article 13-15: https://gdpr-info.eu/art-13-gdpr/]

Regulations and rules: “Right to non-discrimination”



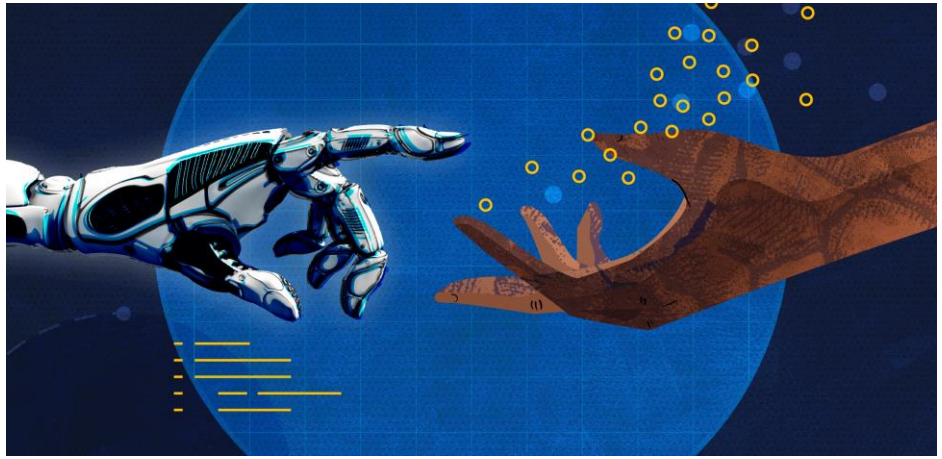
“Any discrimination based on any ground such as sex, race, colour, ethnic or social origin, genetic features, language, religion or belief, political or any other opinion, membership of a national minority, property, birth, disability, age or sexual orientation shall be prohibited.”

[https://ec.europa.eu/info/aid-development-cooperation-fundamental-rights/your-rights-eu/know-your-rights/equality/non-discrimination_en]

Rule/knowledge Extraction

Rules that are usable by humans and helps us better understand different phenomena.

Human in the loop for safety-critical missions



[source: https://medium.com/vsinghbisen/what-is-human-in-the-loop-machine-learning-why-how-used-in-ai-60c7b44eb2c0]

Model Explanations

Scientific Discovery of (governing) rules, cases of flying and drug discovery

- Biology: what visual appearance features a cancerous cell has?
- Biology: how the shapes or features constitute different grades of cancers related to different pathologies (patient outcome and course of treatment)
- Physics: extract governing rules of fluid dynamics
- Environmental studies: extract causal effects
- …



Sustainable progress and method development requires understanding

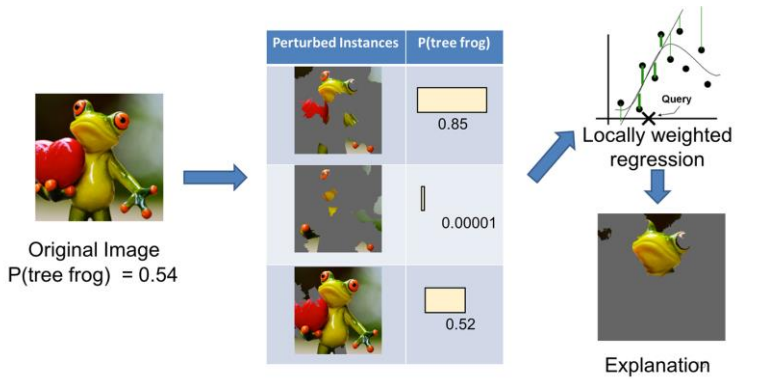
- Learn from mistakes
- Incremental develop better methods inspired by the mistakes

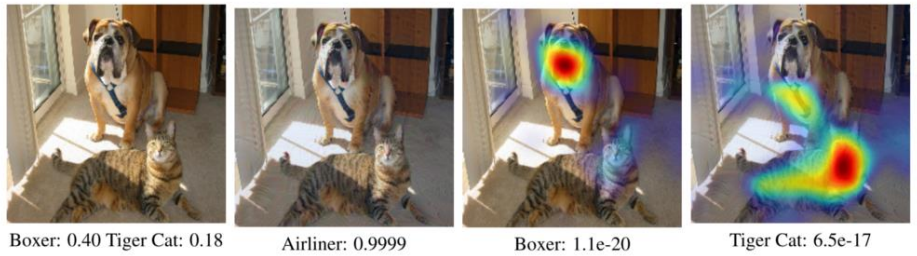


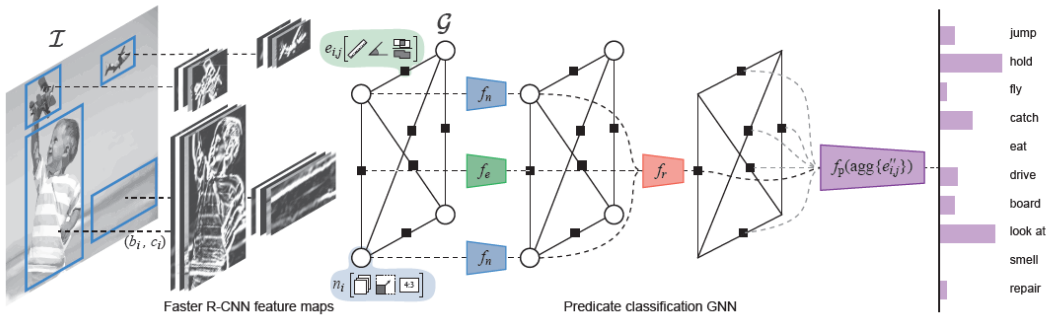
Model Explanations

Types of Explanations

- Explanation by perturbation
- Local Interpretability
- Input Saliency
- Explanation by Attention
- Self-Explanation
- Explainable-by-design methods
- Explanation by Examples
- Black-box Explanations







Model Explanations

Challenges

- Case of persuasiveness
- (post-hoc) Rationalization vs. (ad-hoc) Reasoning
- What are the qualities we look for in explanations?
  - sparsity?
  - faithfullness
  - contrastivity
  - …
- How to measure these qualities?

